

a Data

3 technologies, 20 sections

DLPFC, 10x Visium (12)

STARmap mPFC (3)

MERFISH hypothalamus (5)

expert domain labels

(used only for scoring)

**b Methods**

Deep: HiSTaR, GraphST

Published: BANKSY, SpaGCN

Training-free: linear
diffusion, AnisoST, PCA

3 training seeds each;
one fixed embedding per
section kept for (c)

**c Clustering protocols**

Standard: one mclust-style fit

Robust: best likelihood of 20

Lottery: 20 single random starts

R mclust perturbations:

- permute spot order (10×)
- add 1% noise (10×)

same mixture model (EEE)

**d Evaluation**

ARI vs. labels

seed variance

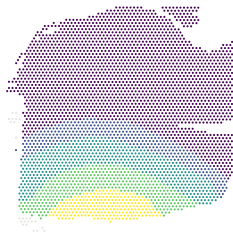
runtime

Wilcoxon tests,
Holm correction

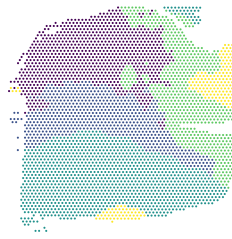
matched vs. standard

e Same fixed HiSTaR embedding (DLPFC 151669), same mixture model, different random start only

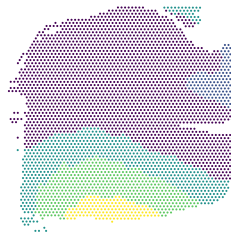
Expert annotation



Random start 9: ARI = 0.23



Random start 13: ARI = 0.67



- Layer 3
- Layer 4
- Layer 5
- Layer 6
- WM

Across 20 random starts on this embedding:
ARI 0.23 - 0.67